

Theorem 1. Let e_I and f_I be supported on $(2n+1)I$. Let $r > 2$ and suppose that

$$\sup_I w(I)^{\frac{1}{2}-\frac{1}{r}} \|e_I\|_{L_r(\mathbb{R}^N, w(x)dx)} \leq 1, \quad \sup_I w(I)^{\frac{1}{2}-\frac{1}{r}} \|f_I\|_{L_r(\mathbb{R}^N, w(x)dx)} \leq 1.$$

If V is an integral operator with integral kernel

$$K : (x, y) \rightarrow \sum_I \lambda_I e_I(x) f_I(y),$$

then

$$\|V\|_{p,\infty} \leq c_{p,n,r} \|\lambda\|_{p,\infty}.$$

Denote $A = \cup_{v \in R} \{x, v\} = \emptyset$.

Theorem 2. Let $M = N + \sum_{v \in R} k(v)$. Let $K : \mathbb{R}^N \times \mathbb{R}^N \rightarrow \mathbb{C}$ be such that

$$|\partial^\alpha K|(x, y) \leq c_{K,\epsilon,\alpha} d(x, y)^{1-M-\alpha}, \quad \text{dist}(x, A), \text{dist}(y, A) \geq \epsilon|x-y|.$$

If V_K is an (unbounded) integral operator with an integral kernel K , then

$$\|M_f V_K\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} \leq c_K \|f\|_{L_2(\mathbb{R}^N)}, \quad p > 2.$$

Lemma 1. Let

$$\Omega = \{(x, y) : x, y \in \Gamma, \quad x \neq y\}.$$

There exists $\mathcal{F} \subset \mathcal{D} \times \mathcal{D}$ such that

- (i) the sets $\{I \times J\}_{I \times J \in \mathcal{F}}$ are pairwise disjoint;
- (ii) $\bigcup_{I \times J \in \mathcal{F}} I \times J = \Omega$;
- (iii) $s(I) = s(J)$ for every $I \times J \in \mathcal{F}$;
- (iv) for every $I \times J \in \mathcal{F}$, we have

$$|x - y|_\infty \in [2s(I), 10s(I)), \quad x \in I, \quad y \in J.$$

- (v) for every $I \times J \in \mathcal{F}$, we have

$$\text{dist}(x, \partial\Gamma), \text{dist}(y, \partial\Gamma) \geq \text{some - constant } s(I).$$

Proof. For each $k \in \mathbb{Z}$, set

$$\Omega_k = \{(x, y) : x, y \in \Gamma, \quad 2^{k+2} \leq \min\{|x - y|_\infty, \text{dist}(x, \partial\Gamma), \text{dist}(y, \partial\Gamma)\} < 2^{k+3}\}.$$

and

$$\mathcal{F}_k = \{I \times J : I, J \in \mathcal{D}, \quad s(I) = s(J) = 2^k, \quad (I \times J) \cap \Omega_k \neq \emptyset\}.$$

Set

$$\mathcal{F}' = \bigcup_{k \in \mathbb{Z}} \mathcal{F}_k.$$

We first show that

$$(1) \quad \Omega = \bigcup_{I \times J \in \mathcal{F}'} I \times J.$$

If $I \times J \in \mathcal{F}_k$ for some given $k \in \mathbb{Z}$, then $s(I) = s(J) = 2^k$ and there exist $x_0 \in I$ and $y_0 \in J$ such that $|x_0 - y_0|_\infty \in [2^{k+2}, 2^{k+3})$. If $x \in I$ and $y \in J$, then $|x - x_0|_\infty < 2^k$ and $|y - y_0|_\infty < 2^k$. By the triangle inequality,

$$2s(I) = 2 \cdot 2^k \leq |x - y|_\infty < 10 \cdot 2^k = 10s(I).$$

In particular, $I \times J \subset \Omega$. If $(x, y) \in \Omega$, then one can uniquely choose $k \in \mathbb{Z}$ such that

$$2^{k+2} \leq |x - y|_\infty < 2^{k+3}.$$

Now one can uniquely choose $I \in \mathcal{D}$ such that $x \in I$ and $s(I) = 2^k$ and $J \in \mathcal{D}$ such that $y \in J$ and $s(J) = 2^k$. Since $(x, y) \in I \times J$ and $(x, y) \in \Omega_k$, it follows that $I \times J \in \mathcal{F}_k$. This gives us (1).

Let $\mathcal{F}$ be the set of maximal elements in $\mathcal{F}'$. We shall show that $\mathcal{F}$ satisfies the desired assertions. It is clear that (iii) and (iv) are satisfied. On the other hand, elements in $\mathcal{F}$ are pairwise disjoint. Indeed, if $Q_1, Q_2 \in \mathcal{F}$ are such that $Q_1 \cap Q_2 \neq \emptyset$, then either $Q_1 \subset Q_2$ or $Q_2 \subset Q_1$. Since both Q_1 and Q_2 are maximal elements of $\mathcal{F}'$, it follows that $Q_1 = Q_2$. We claim that every $Q \in \mathcal{F}'$ is contained in some $R \in \mathcal{F}$. Assuming that this claim has been proved, by (1), we have

$$\Omega = \bigcup_{Q \in \mathcal{F}'} Q = \bigcup_{Q \in \mathcal{F}} Q.$$

Now we turn to prove the claim. If $Q \in \mathcal{F}'$ is maximal, then set $R = Q$, otherwise choose $R \in \mathcal{F}'$ such that $Q \subsetneq R$. For the latter case, we have $s(R) = 2s(Q)$. Indeed, let $Q = I_1 \times J_1$, $R = I_2 \times J_2$. We have $I_1 \subset I_2$ and $J_1 \subset J_2$. Thus,

$$2s(I_2) \leq \inf_{\substack{x \in I_2 \\ y \in J_2}} |x - y|_\infty \leq \inf_{\substack{x \in I_1 \\ y \in J_1}} |x - y|_\infty < 8s(I_1),$$

which implies $s(I_2) < 4s(I_1)$. Since $s(I_1)$ and $s(I_2)$ are powers of 2, it follows that either $s(I_2) = s(I_1)$ or $s(I_2) = 2s(I_1)$. However, by the fact $Q \subsetneq R$, we only have $s(R) = s(I_2) = 2s(I_1) = 2s(Q)$. It remains to show that $R \in \mathcal{F}$. To this end, we check that R is maximal. Actually, if there exists $S \in \mathcal{F}'$ such that $R \subsetneq S$, then, by the preceding argument, $s(S) = 2s(R)$ which further implies $s(S) = 4s(Q)$. However, this contradicts the fact $Q \subsetneq S$ (which implies $s(S) = 2s(Q)$). Hence, R is maximal.

The proof is complete. $\square$

Theorem 3. Integral kernel $K : (x, y) \rightarrow K(x, y)$, $x, y \in \mathbb{R}^N$, of $\Delta^{-\frac{1}{2}}$, where Δ is Dunkl Laplacian is smooth except on the sets $\{x = gy\}$, $g \in G$, or on the planes $\{\langle x, v \rangle = 0\}$, $\{\langle y, v \rangle = 0\}$.

Proof. **IT IS EXACTLY THIS WE CANNOT PROVE YET.** $\square$

Lemma 2. We have

$$K(\lambda x, \lambda y) = \lambda^{1-M} K(x, y), \quad x, y \in \mathbb{R}^N.$$

Proof. Indeed, let D_λ be the dilation by λ . It is obvious (just like in the Euclidean case) that

$$D_\lambda^{-1} T_\xi D_\lambda = \lambda^{-1} T_\xi.$$

Thus,

$$D_\lambda^{-1} \Delta D_\lambda = \lambda^{-2} \Delta.$$

Thus,

$$\Delta^{-\frac{1}{2}} D_\lambda = \lambda D_\lambda \Delta^{-\frac{1}{2}}.$$

In other words,

$$\int_{\mathbb{R}^N} K(x, y) \xi(\lambda^{-1} y) w(y) dy = \lambda \int_{\mathbb{R}^N} K(\lambda^{-1} x, y) \xi(y) w(y) dy.$$

Substituting $y = \lambda z$ in the first integral and $y = z$ in the second one, we write

$$\lambda^{N+\sum_{v \in R} k(v)} \int_{\mathbb{R}^N} K(x, \lambda z) \xi(z) w(z) dz = \lambda \int_{\mathbb{R}^N} K(\lambda^{-1} x, z) \xi(z) w(z) dz.$$

In other words,

$$\lambda^{N+\sum_{v \in R} k(v)} K(x, \lambda z) = \lambda K(\lambda^{-1} x, z).$$

Substituting λx instead of λ and y instead of z , we conclude the argument. $\square$

Lemma 3. *We have (provided that R spans $\mathbb{R}^N$)*

$$f(x)K(x, y) = c \sum_{k, l \in \mathbb{Z}^N} c_{k, l} \sum_{I \in \mathcal{D}} s(I)^{-M} w(I)^{1-\frac{1}{r}} \|f\|_{L_r(I)} e_{k, I}(x) f_{l, I}(y),$$

where $e_{k, I}$ and $f_{l, I}$ are supported in $(2n+1)I$ *can we specify n ?*

$$\sup_{k, I} w(I)^{\frac{1}{2}-\frac{1}{r}} \|e_{k, I}\|_{L_r(\mathbb{R}^N, w(x)dx)} \leq 1,$$

$$\sup_{l, I} w(I)^{\frac{1}{2}-\frac{1}{r}} \|f_{l, I}\|_{L_r(\mathbb{R}^N, w(x)dx)} \leq 1.$$

Proof. Apply Whitney decomposition to the set $(\Gamma \times \Gamma) \setminus \{x \neq y\}$ (we do not even need the linear transformations). Here, each cube $Q \in \mathcal{F}$ is of the shape $I \times J$, where $s(I) = s(J)$. Distance from $x \in I$ to the boundary of Γ is comparable to $s(Q) = s(I)$. Distance from $y \in J$ to the boundary of Γ is comparable to $s(Q) = s(I)$. Distance from $x \in I$ to $y \in J$ is comparable to $s(Q) = s(I)$. Hence, there exists a fixed number n (possibly, dependent on Γ) so that $J \subset (2n+1)I$.

We have (provided that R spans $\mathbb{R}^N$)

$$|\langle x, v \rangle| \geq \epsilon s(I), \quad |\langle y, v \rangle| \geq \epsilon s(I), \quad |x-y| \geq \epsilon s(I), \quad |x|, |y| \leq \epsilon^{-1} s(I), \quad x \in I, \quad y \in J.$$

Set

$$\Omega_\epsilon = \{(x, y) \in \Gamma : |\langle x, v \rangle| \geq \epsilon^2, \quad |\langle y, v \rangle| \geq \epsilon^2, \quad |x-y| \geq \epsilon^2, \quad |x|, |y| \leq 1\}.$$

Thus,

$$\left(\frac{x}{\epsilon^{-1} s(I)}, \frac{y}{\epsilon^{-1} s(I)}\right) \in \Omega_\epsilon, \quad x \in I, \quad y \in J.$$

By Theorem 3, $K \in C^m(\Omega_\epsilon)$ for every $m \in \mathbb{N}$. Hence, there exists a rapidly decreasing sequence $\{c_{k, l}\}_{k, l \in \mathbb{Z}^N}$ such that

$$K(x, y) = \sum_{k, l \in \mathbb{Z}^N} c_{k, l} e^{i\langle k, x \rangle} e^{i\langle l, y \rangle}, \quad (x, y) \in \epsilon^{-1} \Omega_\epsilon.$$

We, therefore, have

$$\begin{aligned} K(x, y) &= (\epsilon^{-1} s(I))^{-M} K\left(\frac{x}{\epsilon^{-1} s(I)}, \frac{y}{\epsilon^{-1} s(I)}\right) = \\ &= (\epsilon^{-1} s(I))^{-M} \sum_{k, l \in \mathbb{Z}^N} c_{k, l} e^{\frac{i\epsilon\langle k, x \rangle}{s(I)}} e^{\frac{i\epsilon\langle l, y \rangle}{s(I)}}, \quad x \in I, \quad y \in J. \end{aligned}$$

Set $J_I = \cup_{J \times J \in \mathcal{F}} J$. Integral kernel of $M_f \Delta^{-\frac{1}{2}}$ is

$$\sum_{k, l \in \mathbb{Z}^N} c_{k, l} \sum_{I \in \mathcal{D}} (\epsilon^{-1} s(I))^{-M} f(x) e^{\frac{i\epsilon\langle k, x \rangle}{s(I)}} \chi_I(x) \cdot e^{\frac{i\epsilon\langle l, y \rangle}{s(I)}} \chi_{J_I}(y).$$

Set

$$e_{k, I}(x) = \frac{w(I)^{\frac{1}{r}-\frac{1}{2}}}{\|f\|_{L_r(I, w(x)dx)}} f(x) e^{\frac{i\epsilon\langle k, x \rangle}{s(I)}} \chi_I(x).$$

Obviously,

$$\sup_{k, I} w(I)^{\frac{1}{2}-\frac{1}{r}} \|e_{k, I}\|_{L_r(\mathbb{R}^N, w(x)dx)} = 1.$$

Set

$$f_{l,I}(x) = w(I)^{-\frac{1}{2}} e^{\frac{i\epsilon\langle l,x \rangle}{s(I)}} \chi_{J_I}(x).$$

Obviously,

$$\begin{aligned} \sup_{l,I} w(I)^{\frac{1}{2}-\frac{1}{r}} \|f_{l,I}\|_{L_r(\mathbb{R}^N, w(x)dx)} &= \sup_I w(I)^{-\frac{1}{r}} \|\chi_{A_I}\|_{L_r(\mathbb{R}^N, w(x)dx)} \leq \\ &\leq \sup_I w(I)^{-\frac{1}{r}} \|\chi_{(2n+1)I}\|_{L_r(\mathbb{R}^N, w(x)dx)} = \left(\sup_I \frac{w((2n+1)I)}{w(I)} \right)^{\frac{1}{r}} = c_n < \infty. \end{aligned}$$

Integral kernel of $M_f \Delta^{-\frac{1}{2}}$ is

$$\sum_{k,l \in \mathbb{Z}^N} c_{k,l} \sum_{I \in \mathcal{D}} (\epsilon^{-1} s(I))^{-M} w(I)^{1-\frac{1}{r}} \|f\|_{L_r(I, w(x)dx)} e_{k,I}(x) f_{l,I}(y).$$

□

By Theorem 1 and by the triangle inequality, the norm $\|M_f \Delta^{-\frac{1}{2}}\|_{N,\infty}$ is controlled by

$$\sum_{k,l \in \mathbb{Z}^N} |c_{k,l}| \cdot \left\| \left\{ s(I)^{-M} w(I)^{1-\frac{1}{r}} \|f\|_{L_r(I, w(x)dx)} \right\}_{I \in \mathcal{D}} \right\|_{N,\infty}.$$

Note that

$$w(x) \sim s(I)^{\sum_{v \in R} k(v)}, \quad x \in I.$$

Thus,

$$w(I) = \int_I w(x)dx \sim s(I)^{N + \sum_{v \in R} k(v)},$$

$$\|f\|_{L_r(I, w(x)dx)} \sim s(I)^{\frac{1}{r} \sum_{v \in R} k(v)} \|f\|_{L_r(I, dx)}.$$

So, $\|M_f \Delta^{-\frac{1}{2}}\|_{N,\infty}$ is controlled by

$$\begin{aligned} &\left\| \left\{ s(I)^{-M} s(I)^{(1-\frac{1}{r})(N + \sum_{v \in R} k(v))} s(I)^{\frac{1}{r} \sum_{v \in R} k(v)} \|f\|_{L_r(I, dx)} \right\}_{I \in \mathcal{D}} \right\|_{N,\infty} = \\ &= \left\| \left\{ s(I)^{-M} s(I)^{N + \sum_{v \in R} k(v)} s(I)^{-\frac{N}{r}} \|f\|_{L_r(I, dx)} \right\}_{I \in \mathcal{D}} \right\|_{N,\infty}. \end{aligned}$$

Since

$$M = N - 1 + \sum_{v \in R} k(v),$$

it follows that the latter expression is

$$\left\| \left\{ s(I)^{1-\frac{N}{r}} \|f\|_{L_r(I, dx)} \right\}_{I \in \mathcal{D}} \right\|_{N,\infty}.$$

By Zhijie lemma, this is controlled by $\|f\|_{L_d(\mathbb{R}^N)}$.